signal pattern of spoken words. When there is a match, that sound is transmitted onto the other end of the line.

Applications

Applications are huge in number. The major one being able to make silent calls even though you are standing amidst thousands without disturbing others. Now you can avoid unwanted noises around you to reach out to the person you're talking to.

This technology is very useful at areas where audible speech is impossible: astronauts in space (remember sound does not travel in vacuum), divers, fighter pilots, or those working in nasty environments can communicate successfully.

Even the speech impaired can talk over the phone now. This technology is helpful for people without vocal cords or those who are suffering from aphasia, an acquired language disorder that causes difficulty in reproducing or comprehending spoken or written language. Such people would largely benefit from this technology.

The whole range of information security issues will take a positive bend where information can be exchanged vocally without any unwanted interception. To tell a secret PIN number or credit card number on the phone would now be easy, as there's no one to eavesdrop any more.

Since the electrical signals generated are universal, they can be translated into any language. Native speakers can speak in their language and language translators can translate it before sending it to the other side. Hence, it can be converted to any language of choice currently being German, English and French.


The silent sound setup in use

Restrictions

Translation into majority of the languages works well. But for languages such as Chinese, different tones hold different meanings, facial movements being the same. Hence this technology is difficult to apply in such situations. From the security point of view, recognising who you're talking to gets complicated.

Even differentiating between people and emotions can't be done. This means you'll always feel you're talking to a robot. This device presently needs nine leads to be attached to your face which is quite impractical to make it usable.

Future prospects

Silent sound technology gives way to a bright future to speech recognition technology. From simple voice commands to memorandums dictated over the phone, all this is fairly possible in noisy public places.

Also without having electrodes hanging all around your face, these electrodes will be incorporated into cellphones making them look more usable.

Some suggest it would be great to have lip reading based on image recognition and processing rather than using electromyography. Also, incorporarting nano technology would be a mentionable step towards making the device handy.


Hopefully silent speech technology could be a commercial product soon

Conclusion

The engineers claim that the device is working with 99 per cent efficiency. This means out of 100 words being recognised, one is interpreted wrong. The developers say the device would be everyday use in about five to 10 years.